KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · MATH

Malik's Mural Project

Name: _____ Date: _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Before the First Tile

Malik's art class is about to start a giant mural project. Before you meet him, warm up on the exact math skills he'll need today.

1 Match each word to its meaning. Draw a line, or write the matching letter next to each word.

___ **standard algorithm**	**A.** A smart, quick guess made by rounding first.
___ **partial product**	**B.** The stacked, step-by-step way of multiplying larger numbers on paper.
___ **regroup**	**C.** A number being multiplied.
___ **product**	**D.** Carrying extra value into the next place value column.
___ **factor**	**E.** One piece of a multiplication problem, found before adding all the pieces together.
___ **estimate**	**F.** The answer to a multiplication problem.

2 Preview: multiply 14 × 3 using partial products. Break 14 into 10 + 4, multiply each part by 3, then add. Show every step.

3 Have you ever helped with a big group project, like a mural, poster, or art display? Write 1–2 sentences about it.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Counting Every Tile

Great work on today's lesson! Now show off what you learned about the standard algorithm, partial products, and regrouping.

1. The gym mural was such a hit that Malik's class is designing a SECOND mural for the cafeteria wall: 32 tiles across and 26 tiles down. Use the standard algorithm to find how many tiles it needs. Show every step, including both partial products.

2. Each jar of paint covers 148 tiles. Malik has 6 jars. How many tiles can all 6 jars cover? Use the standard algorithm and show where you regroup. Is that enough paint for the cafeteria mural?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON · CONTINUED

Counting Every Tile

3 Gold border tiles come in packs of 28, and the class needs 19 packs. First, ESTIMATE the total number of gold tiles by rounding each factor to the nearest ten. Then use the standard algorithm to find the exact product. Was your estimate higher or lower than the exact answer? Why?

4 Malik's classmate multiplied 24 × 18 and got 216. Malik checked his work: the first partial product (24 × 8 = 192) was right, but the second one was written as 24 instead of 240. Explain in 1–2 sentences what mistake his classmate made, and find the correct product.

★ **Challenge:** Design your own tile mural! Choose a 2-digit number of tiles across and a 2-digit number of tiles down. Use the standard algorithm to find the total number of tiles, then estimate to check that your answer makes sense.

Looking Ahead: Next week we divide BIG numbers using 2-digit divisors — Deja needs help splitting her class's food drive cans evenly among 24 pantries! Bring this sheet back to your next lesson.